

# Physical Chemistry (I)

## Lecture 16

2차 중간고사 안내 및 복습



# 2차 중간고사 안내

- 시간: 5월 9일(월) 오후 3:00~4:00
- 장소: 화학관 315호(홀수 학번), 414호(짝수 학번)
- 계산기 불필요

# 시험 범위

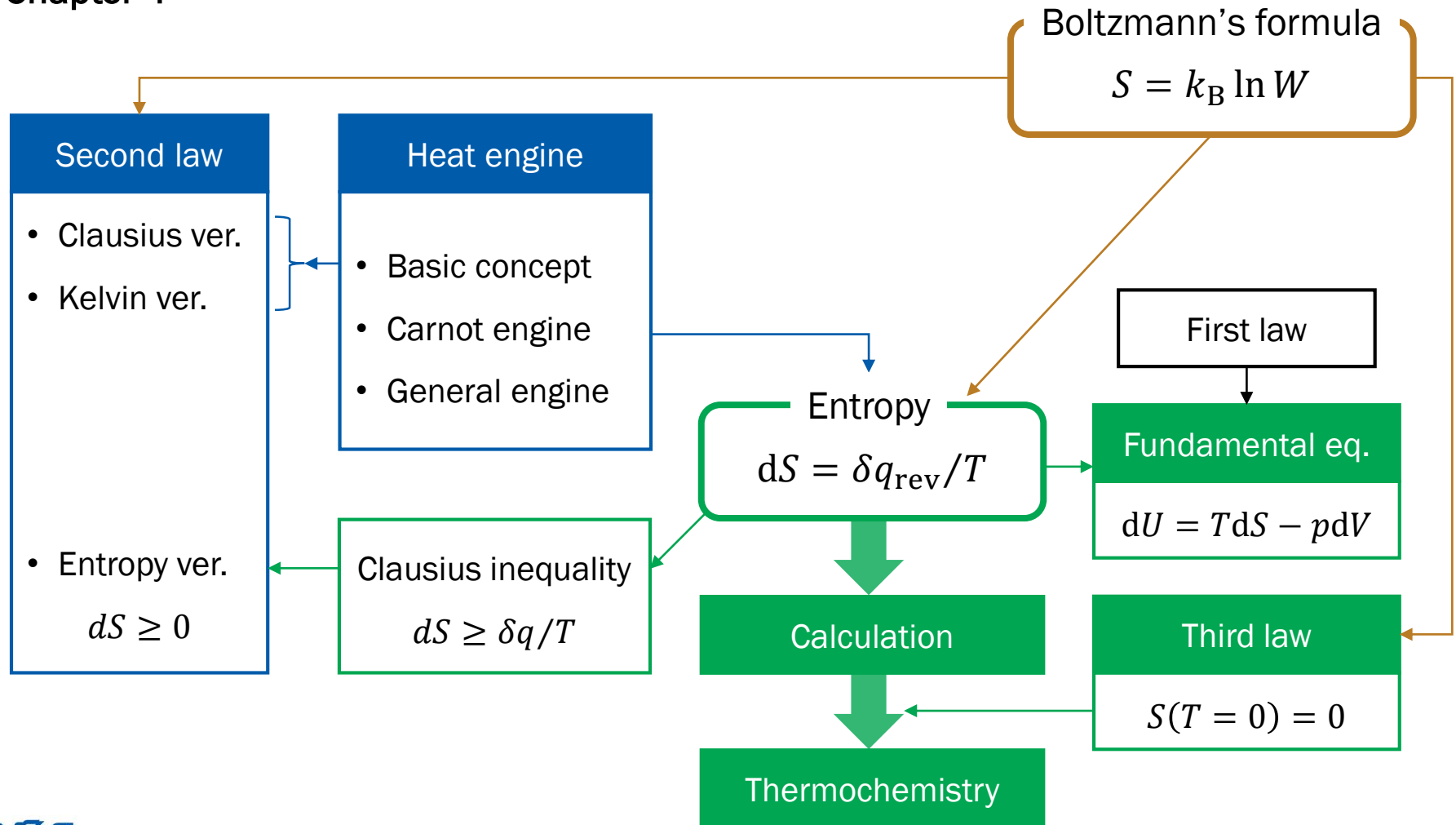
- 4장 "엔트로피"
- 5장 "헬름홀츠 및 기브스 에너지"
- 6장 "상 전이"
- 7장 "단순 혼합물" 진도 나간 곳까지

# 시험 문제

## 총 일곱 문제

- 과제 문제 혹은 그 변형(3문제)
  - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 개념 설명 및 공식 유도(4문제)
  - 강의 자료 기준

## Chapter 4



# Concepts and Derivations in Chpt 4

- [C, D] Entropy
  - Thermodynamic definition
  - Entropy change of surroundings
- [C, D] Clausius inequality
  - Entropy version of the second law
  - Entropy and spontaneity
- [C] Fundamental equation of thermodynamics

# Concepts and Derivations in Chpt 4

- [D] Entropy calculations
  - Heat transfer
  - Expansion
  - Phase transition
  - Heating
- [C] Nernst heat theorem and the third law
- [C] Thermochemistry of entropy
  - Standard entropy
  - Standard reaction entropy



# Concepts and Derivations in Chpt 4

- [C] Entropy as an index of exhaustion
- [C] Boltzmann's formula
- [D] Applications of Boltzmann's formula
- [C] Reinterpretations of the laws
- [C, D] Independent spin model



## Chapter 5

### Thermodynamic potentials

Clausius ineq.

First law

Internal energy

$$U$$

Helmholtz energy

$$A = U - TS$$

Enthalpy

$$H = U + pV$$

Gibbs energy

$$G = H - TS$$

Spontaneity

$$\Delta A_{T,V} \leq 0$$

$$\Delta G_{T,p} \leq 0$$

Maximum work

$$A_T = w_{\max}$$

$$G_{T,p} = w_{\text{add,max}}$$

Differential forms

Applications

- Gibbs-Helmholtz equation
- Maxwell relations
- Thermodynamic equation of state

Thermochemistry

Pressure dependence

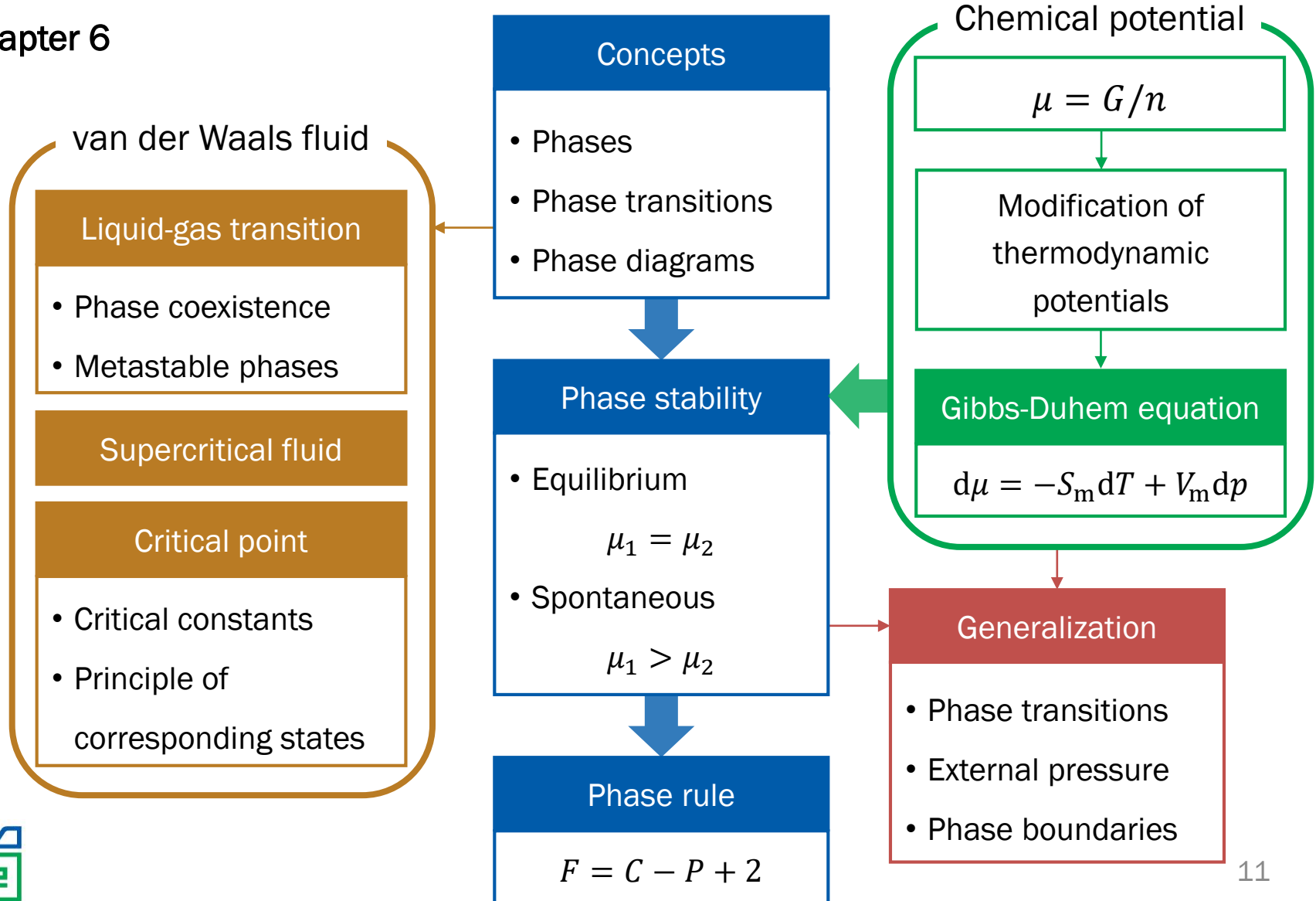


# Concepts and Derivations in Chpt 5

- [C] Helmholtz and Gibbs energies
- [D] Spontaneity: Clausius inequality
- [D] Meaning of Helmholtz and Gibbs energies
- [D] Thermodynamic potentials
- [D] Gibbs-Helmholtz equation
- [D] Maxwell relations
- [D] Thermodynamic equation of state
- [D] Pressure dependence of Gibbs energy



## Chapter 6



# Concepts and Derivations in Chpt 6

- [C] Phases and phase transitions
- [C, D] Chemical potential
- [D] Rewriting thermodynamic potentials
- [D] Gibbs-Duhem equation
- [D] Phase transition and chemical potential
- [C] Phase diagrams
- [C, D] Phase rule
- [C] Liquid-gas transitions



# Concepts and Derivations in Chpt 6

- [C] vdW: Liquid-gas transitions
  - [C] Maxwell construction
  - [D] Lever rule
  - [C] Binodal and spinodal lines
- [C] vdW: Supercritical fluid
- [C] vdW: Critical point
  - [D] Critical constants
  - [D] Critical compression factor
  - [D] Reduced variables
  - [C] Principle of corresponding states



# Concepts and Derivations in Chpt 6

- [C] Chemical potentials at transition
  - [D] Use of Gibbs-Duhem equation
- [D] Pressure effect on liquid-gas equilibrium
- [C] Phase boundaries
  - [D] Clausius equation
  - [D] Clausius-Clapeyron equation
  - [C] Interpretations

# Concepts and Derivations in Chpt 7

- Extension to multi-component systems
  - [C] Thermodynamic potentials
  - [C, D] Partial molar quantities
  - [C, D] Chemical potential
  - [D] Gibbs-Duhem equation